

RDP Brute-Force Attack Investigation Report

Executive Summary

This report documents the investigation of suspicious Remote Desktop Protocol (RDP) authentication activity identified through Splunk log analysis. The investigation focused on detecting brute-force login attempts, analyzing attacker behavior, identifying targeted accounts, and determining whether unauthorized access was achieved.

The analysis revealed multiple failed authentication attempts originating from the same source IP address, followed by a successful login event. The activity is consistent with a brute-force attack and indicates a potential account compromise.

Incident Details

Incident Type

RDP Brute-Force Attack

Severity

High

Detection Method

Splunk Security Monitoring

Data Source

Windows Authentication Logs

Investigation Date

May 2026

Objective

The purpose of this investigation was to:

- Identify failed RDP authentication attempts
- Detect brute-force attack patterns
- Analyze source IP activity
- Investigate targeted user accounts
- Determine whether a successful compromise occurred

- Document findings and recommendations

Environment Overview

The dataset consisted of Windows authentication logs containing:

- Event ID 4625 (Failed Login)
- Event ID 4624 (Successful Login)
- Logon Type 10 (Remote Desktop Protocol)

These events were ingested into Splunk for security monitoring and analysis.

Investigation Methodology

Step 1: Log Collection

Authentication logs were uploaded into Splunk and indexed under a dedicated data source for analysis.

Step 2: Failed Login Analysis

Failed authentication events were reviewed to identify suspicious login attempts targeting RDP services.

Query Used:

```
index=rdp eventcode=4625 logon_type=10
```

The analysis revealed multiple authentication failures originating from the same source IP address.

Step 3: Source IP Investigation

Source IP addresses were analyzed to identify systems generating excessive failed login attempts.

Query Used:

```
index=rdp eventcode=4625  
| stats count by src_ip  
| sort - count
```

The investigation identified one IP address responsible for a significantly higher number of failed authentication attempts than other systems.

Step 4: User Account Analysis

Targeted user accounts were reviewed to determine whether administrative accounts were being attacked.

Query Used:

```
index=rdp eventcode=4625  
| stats count by user  
| sort - count
```

The results indicated repeated login attempts against privileged accounts.

Step 5: Brute-Force Detection

A threshold-based analysis was performed to identify potential brute-force activity.

Query Used:

```
index=rdp eventcode=4625  
| stats count by src_ip  
| where count > 5
```

The identified IP address exceeded the defined threshold and was classified as suspicious.

Step 6: Successful Login Correlation

Authentication events were correlated to determine whether a successful login occurred after repeated failures.

Query Used:

```
index=rdp (eventcode=4624 OR eventcode=4625)  
| stats values(eventcode) as codes count by src_ip user
```

The investigation revealed that multiple failed login attempts were followed by a successful authentication event from the same source IP address.

This behavior is commonly associated with successful brute-force attacks.

Findings

The investigation identified the following:

- Multiple failed RDP authentication attempts from a single source IP
- Repeated targeting of user accounts
- Brute-force attack patterns
- Successful login following repeated authentication failures
- Potential unauthorized access to a system

The observed behavior aligns with known credential attack techniques.

Indicators of Compromise (IOCs)

IOC Type	Value
Attack Type	RDP Brute Force
Event ID	4625
Event ID	4624
Logon Type	10
Activity	Repeated Authentication Failures
Activity	Successful Login After Failures

MITRE ATT&CK Mapping

Technique	MITRE ID
Brute Force	T1110
Remote Services: RDP	T1021.001
Valid Accounts	T1078

Impact Assessment

If the successful login was unauthorized, the attacker may have gained access to the target system using valid credentials.

Potential risks include:

- Unauthorized system access
- Privilege escalation

- Lateral movement
- Data theft
- Persistence within the environment

Recommendations

1. Enable Multi-Factor Authentication (MFA) for all remote access services.
2. Restrict RDP access to trusted networks and VPN users.
3. Implement account lockout policies after multiple failed login attempts.
4. Continuously monitor Windows authentication events.
5. Investigate suspicious source IP addresses.
6. Review privileged account activity for signs of misuse.
7. Deploy automated alerts for excessive failed authentication attempts.

Conclusion

The investigation confirmed suspicious RDP authentication activity consistent with a brute-force attack. Multiple failed login attempts followed by a successful authentication event indicate a potential compromise scenario. Immediate review of affected accounts and implementation of preventive controls are recommended to reduce the risk of future attacks.

Analyst

Muhammad Talha

SOC Analyst (Aspiring) | Splunk Enthusiast | Cybersecurity Student